The empirical recurrence for OEIS A257380: recovering a conjecture that is not written down

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Abstract

OEIS A257380 records "Empirical recurrence of order 81", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 92$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 81, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 81 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A257380 is "Number of (n+2)X(1+2) 0..2 arrays with no 3x3 subblock diagonal sum equal to the antidiagonal sum or central row sum less than the central column sum.". It has offset 1 and begins

$$9300, 119716, 1719620, 24720784, 349003536, 4925793856, 69733153264, 987300576900, \ \dots$$

The entry states:

> Empirical recurrence of order 81 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 15:22:20, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 757$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 92$ of the 757, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 92$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1514$ exact terms of the model returns order 81 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{81} c_i\, a(n-i),$$

c_1	5	c_{22}	84637945352748	c_{43}	885362227756793 40544	c_{64}	−5811897867583419
c_2	0	c_{23}	236542237585124	c_{44}	3565072963717893 7344	c_{65}	−1862105235649441
c_3	850	c_{24}	−1591075764064904	c_{45}	−20398293164657 2040192	c_{66}	−3419968156860206
c_4	9400	c_{25}	−4668038304244294	c_{46}	−26071952036088 8008704	c_{67}	2659951449177325
c_5	43938	c_{26}	4207045554359794	c_{47}	−64782175238218 6450944	c_{68}	−672496960463553
c_6	267810	c_{27}	21040255729867570	c_{48}	−67028460997286 4790528	c_{69}	1270480181465081
c_7	463689	c_{28}	11006262278042519	c_{49}	28623357819906 424832	c_{70}	2094671575641174
c_8	−3374045	c_{29}	15063137020669131	c_{50}	−11983688330433 1870208	c_{71}	−102237824138765
c_9	−11048096	c_{30}	56309550170504916	c_{51}	10005469390390 19433984	c_{72}	458862520514379
c_{10}	−4341732	c_{31}	49632689235835430	c_{52}	11372825986357 53701376	c_{73}	435842011203895
c_{11}	41999778	c_{32}	2258611102275450	c_{53}	73178072397186 0987904	c_{74}	460061613535893
c_{12}	920656046	c_{33}	−137866219995795324	c_{54}	12050381766977 91545344	c_{75}	−27813685940872
c_{13}	−774928762	c_{34}	−1296555999014782784	c_{55}	25201849629735 5968512	c_{76}	195259191344182
c_{14}	−50728144354	c_{35}	−4089219051466749019	c_{56}	50552121275502 6116608	c_{77}	−66906601964122
c_{15}	−119780815888	c_{36}	−2892789005672415975	c_{57}	−27993077102772 9055744	c_{78}	−89227567617277
c_{16}	387829962456	c_{37}	−2061785192944299 52	c_{58}	−25047910325250 5821184	c_{79}	0
c_{17}	1527691787060	c_{38}	4678950531345410 56	c_{59}	−79249861575747 3579008	c_{80}	−54043195528445
c_{18}	2138600636178	c_{39}	1700225680469785 0304	c_{60}	−10274126326142 45236736	c_{81}	180143985094819
c_{19}	2752275088818	c_{40}	2797753546627112 2432	c_{61}	−53980188241712 6440960		
c_{20}	−16799160458018	c_{41}	4549566055548456 4800	c_{62}	−96041086142499 9399424		
c_{21}	−55585329540450	c_{42}	7059300870824474 4768	c_{63}	−29948094848463 9555584		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 81, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $81 + 4$ terms removed returns order 81 as well, so $\deg q_{\min} = 81 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $81 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 81 that this sequence satisfies — and that family turns out to have exactly one member.

References

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